C-7 Ventricular Fibrillation/Pulseless Ventricular Tachycardia
Adult and Pediatric
Effective October 3, 2024
Revised: July 2025

CPR performed in accordance with AHA guidelines. LUCAS device application is at the discretion of the paramedic if it's applied prior to packaging the patient for transport.

Adults without ROSC should not be moved without LUCAS in place unless the current location prohibits placement of the LUCAS device.

Removal of LSB or scoop stretcher is not necessary if transporting an arrest victim.

If pulse is ever palpable at any point immediately load and go and transport. Follow ROSC protocol. Unconscious patients should be transported with LUCAS device in place if feasible. If necessary, the compressor piston may be removed but back plate should remain in position.

It is preferred to administer medications following the rhythm check with resumption of CPR. Follow administration of medications with a 20 ml fluid bolus in the adult and a 5-10ml flush in the pediatric patient.

Follow appropriate ACLS algorithm dictated by monitor check and patient assessment during any cardiac arrest.
Maternal Arrest - Treat mother per appropriate protocol with immediate notification to Medical Control and rapid
Transport. Place mother supine and perform Manual Left Uterine Displacement moving uterus to the patient's left side.

Defibrillate initial ventricular fibrillation/pulseless ventricular tachycardia at 360 joules biphasic. Repeat defibrillation attempts for VF/VT at 2 minute CPR cycles if indicated. Consider changing pad placement for refractory VF/VT.

If basic airway with BVM is adequate, defer advanced airway until after first administration of epinephrine. Exception: multiple providers on scene. Follow Airway protocol. Monitor capnography once definitive airway placed. Maintain capnography above 10 mmHg.

Establish an IV/IO and run normal saline at KVO.

Administer **epinephrine 1:10 concentration** 1 mg IV or IO. Repeat every 3- 5 minutes prn cardiac arrest.

Following initial epinephrine, administer **Amiodarone 300 mg IV or IO** if patient continues in VF or pulseless VT at next rhythm check. This may be **repeated in 3-5 minutes at a dose of 150 mg IV or IO.**

Continued on next page

Omit amiodarone in cardiac arrest due to cocaine/methamphetamine/TCA intoxication. Administer lidocaine 1 mg/kg IV/IO. May repeat at 0.5 mg/kg every five minutes to a maximum of 3 mg/kg.

In some cases, high quality CPR may induce consciousness. Generally, CPR induced consciousness starts after a run of high-quality compressions and dissipates rapidly after cessation of CPR. If the patient demonstrates any signs of awareness during cardiac arrest that interferes with CPR:

Administer Ketamine 1mg/kg IV or IO

Defibrillate initial ventricular fibrillation/pulseless ventricular tachycardia at 2j/Kg biphasic. Escalate dosing of defibrillation at subsequent monitor check of VF or pulseless VT to 4j/Kg joules biphasic. Dose may be escalated in a stepwise fashion to a maximum of 10j/KG. Utilize pediatric pads on the LP15 for pediatric patients< 15Kg or 1 year old.

If basic airway with BVM is adequate, defer advanced airway until after first administration of epinephrine. Exception: multiple providers on scene. Follow Airway protocol. Monitor capnography once definitive airway placed. Maintain capnography above 10 mmHg.

Establish an IV/IO and run normal saline at KVO

Administer **epinephrine 1:10,000 concentration** 0.01mg/Kg or 0.1ml/Kg IV or IO. Repeat every 3- 5 minutes prn cardiac arrest.

Following initial epinephrine, administer **Amiodarone 5mg/Kg IV or IO** if patient continues in VF or pulseless VT at next rhythm check. This may be **repeated in 3-5 minutes at a dose of 5mg/Kg IV or IO.**

Omit amiodarone in cardiac arrest due to cocaine/methamphetamine/TCA intoxication. Administer lidocaine 1 mg/kg IV/IO. May repeat at 0.5 mg/kg every five minutes to a maximum of 3 mg/kg.

CONFIRM DOSING WITH LENGTH BASED TAPE

Consider Reversible Causes:

HYPERKALEMIA: (History of missed dialysis, ingestion of potassium, crush syndrome)
Administer **1 G (1 amp) 10% Calcium Chloride IVP/IO**

Flush line prior to administration of Sodium Bicarbonate OR Epinephrine

Administer **8.4% Sodium Bicarbonate 1mEq/kg IVP**/IO **kg (Not to exceed 100 mEq**)

Flush Line prior to administration of Epinephrine

Administer 20 mg/kg 10% Calcium Chloride IVP/IO

8.4% Sodium Bicarbonate 1 mEq/kg IVP/IO to a maximum dose of 50 mEq.

- Utilize 4.2% concentration patients under 1 yr old

Hypoglycemia:
If glucose is below 60 mg/dL, administer 25 grams of **dextrose** to the adult IV/IO. Push early in the resuscitation.

If ≤ one month old: **Give D10 5 mL/kg IV**
1 month to 2 Years: **D25 2 mL/kg IV initial dose, give D10 as above**.
> 2 Years: **D50 1 mL/kg IV Maximum 25 g per dose or give D10 as above.**

HYPOVOLEMIA:
If lungs are dry, run IV wide open for a fluid bolus of 500 mL normal saline, repeat as needed.

In a pediatric patient not exhibiting signs of pubescence, use Buretrol tubing or attach a syringe and stopcock device to administer rapid IV fluid boluses.

Administer a 20 ml/kg normal saline fluid bolus. Reassess.

OVERDOSE:

- In the event of *TCA overdose or cocaine intoxication*, flush the IV/IO line and administer **sodium bicarbonate 1 mEq/kg (Not to exceed 100 mEq**) IV/IO push. Circulate the sodium bicarbonate with CPR for two minutes and evaluate response. Flush IV line.
- In the event of *TCA overdose*, hyperventilate the patient at a rate of 20 and administer IVF 500 ml bolus.
- In a *beta blocker/calcium channel blocker overdose*: Administer **1 G (1 amp) 10% Calcium Chloride IVP/IO. Flush IV line.**

- **In the event of TCA overdose, Administer bicarbonate 1 mEq/kg** IV/IO push. Circulate the sodium bicarbonate with CPR for two minutes and evaluate response. Flush IV line.
    - hyperventilate the infant at a rate of 30, a child at a rate of 25 and administer a 20ml/Kg fluid bolus.
- In a *beta blocker/calcium channel blocker overdose*: Administer **Calcium Chloride 10% solution at 0.2 ml/Kg IV/IO. Flush IV line.**